Keep the Referent: Anchor-Preserving Token Compression for Privacy-Aware Inference on IoT Logs

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Paper under double-blind review

***Abstract*—Long IoT device logs are expensive to feed to a local language model and are dense with identifiers that must not leave the device. A popular remedy is token compression driven by *redundancy*: drop what repeats. We show that this signal is anti-correlated with the signal a privacy-aware system actually needs. On a benchmark built from six public datasets, redundancy-based compression deletes precisely the entity a question asks about: in all 40 multi-entity cases the answer string survives in the compressed context while the *referent* does not, and the model answers UNKNOWN in 39 of them, yielding F1 $= 0.000$. The same signature appears in a published masking-based pipeline (`mask_quito_x`: F1 $= 0.000$, gold span present in 70% of cases), which locates the failure in the problem formulation rather than in any single implementation. We present PRIVTRIM, a training-free pre-processing layer that decides per unit whether to delete, alias, merge or keep, protects the units containing the queried entity, and applies its alias map to the question as well as the context. At a 50% token budget PRIVTRIM reaches F1 $= 0.745$ versus 0.719 uncompressed, with *zero* prompt-level leakage of the protected value (the uncompressed prompt leaks in $112/112$ cases; the strongest purely compressive baseline in $44/112$), and it ranks first in every stratum we can stratify by—answer position (3/3) and entity count (2/2)—including the tail regime where trimming hurts most (0.833 vs. 0.667). The advantage holds on a second model, across five budget levels down to 10%, and across two mechanical question rephrasings.**

***Index Terms*—privacy-preserving inference, token compression, IoT logs, local language models, entity preservation, benchmark.**

I. INTRODUCTION

Operational logs from IoT deployments are long, repetitive, and full of identifiers: device IDs, account numbers, MAC addresses, session tokens. Two constraints follow. First, context length makes local inference expensive or impossible—the longest case in our benchmark is 18,608 tokens, which a single 32 GB accelerator can barely hold. Second, the identifiers are exactly the data that regulation and common sense say should not be shipped to a model. Both constraints push toward the same intervention: compress the context before inference.

The dominant family of methods for this task scores tokens, sentences or chunks by *redundancy* and keeps the least redundant ones. The intuition is sound for summarisation, retrieval and document QA, where the goal is to preserve gist. It fails catastrophically, however, when the question is *about one of the repeated entities*. A device identifier that appears fifteen times is, by any information-theoretic or learned redundancy measure, among the cheapest tokens to discard—and it is the one token the answer depends on. Removing it does not merely lower the score; it removes the referent, so the model cannot resolve the question at all.

a) What we show.: (i) On a benchmark assembled from six public datasets (§III), redundancy-based compression yields F1 $= 0.000$ on multi-entity cases while leaving the answer string in the context—the failure is referential, not informational. (ii) The same failure signature appears in a published masking pipeline, so it characterises the problem class, not one implementation. (iii) A training-free decision layer that *protects* the units holding the queried entity, and that generalises the surrounding identifiers via an alias map applied to the question as well as the context, removes the failure: F1 $= 0.745$ at 46.6% of the tokens with zero prompt-level leakage. (iv) The advantage is not an artefact of one stratum: PRIVTRIM ranks first in every answer-position and entity-count stratum we can form, on both models tested, and grows as the budget tightens.

b) Contributions.:

- We identify and *measure* the cell that prior benchmark designs cannot express: the queried object is also the object to protect. On three independent checks the published batch-1 benchmark has no overlap between queryable and sensitive objects ($0/96$), which explains why the literature reports compression as harmless.
- We give a training-free, model-agnostic decision layer (four-quadrant routing, anchor protection, question-side aliasing, hard budget accounting) that is auditable: no new training, deterministic, and bit-reproducible within a configuration.
- We report a reusable benchmark of 112 cases drawn from six public datasets with deterministically derived gold answers, together with six published compressors re-run on it, per-source and per-stratum breakdowns, and an evidence audit that recomputes all 184 reported numbers from raw artefacts.

II. RELATED WORK AND WHY THE LITERATURE CONCLUDES OTHERWISE

a) Token compression.: LLMLingua-2 trains a token classifier to drop low-information tokens [1]; Provence performs context pruning jointly with question relevance [2]; CPC uses an information-bottleneck objective [3]; Quito-X exploits attention attribution [4]. All four are evaluated on document

QA and summarisation, where the answer is a span of ordinary text. PRIVTRIM is complementary: it does not learn a better redundancy score, it overrides the decision for units whose referential role is load-bearing.

b) PII redaction and privacy in retrieval.: De-identification pipelines replace detected PII with typed placeholders before the model sees the text. When such redaction is combined with compression (as in `mask_quito_x`), each component behaves as designed—and the composition deletes the query target. Our results show the qualitative difference between *masking* (replace the value, keep a referential handle) and *masking plus compression* (lose the handle as well).

c) Why the literature reports no harm.: In the published batch-1 benchmark that motivated our work, queryable objects and protected objects are disjoint: three checks (question mentions a canary, answer contains a canary, canary appears in the queried passage) all return $0/96$. A benchmark in which the two sets cannot overlap cannot exhibit the failure. Re-running `mask_llmlingua2` on that benchmark degrades 11% relative to uncompressed, whereas on our multi-entity cases it collapses to 0.000. We therefore frame our result not as a refutation but as measurement of a cell that was previously not representable.

III. PROBLEM SETTING AND BENCHMARK

a) Threat and task model.: A local model with $\leq$ 7B parameters answers a single question over one long log excerpt. The excerpt contains k entities of the same type (device, account, user) and a set of sensitive attribute values. The question names an entity and asks for one of its attributes. The system must (a) keep inference within a token budget and (b) ensure that no protected value is recoverable from what is sent to the model, including from the question itself.

b) Data.: Cases are built from six public datasets (Table I): SmartBench TDCS (device-context anomaly detection, closest to IoT logs), PEC-Home (multi-user home preferences), SimuHome (device-control tasks), PrivacyLens (a privacy-specific benchmark), and two tool-memory conversation corpora. Context length ranges from 1,979 to 18,608 Qwen tokens; questions use 71 distinct templates over 30 attribute keys. Sensitive values are *synthetic* canaries injected per case (public data cannot carry real PII); the log content around them is the datasets' own records, and the case file carries a SHA-256 per case. Gold answers are derived deterministically from the source structured records, never from a model.

c) Compression arms.: All arms share the same budget accounting: per case the target is $B = \mathrm{round}(r \cdot n_{\mathrm{full}})$ tokens, and any arm whose realised context exceeds B is a hard error, so a run either satisfies the budget or fails loudly. We compare (i) `full`, the uncompressed prompt; (ii) `budget_cluster`, redundancy-based unit selection; (iii) `budget_cluster_link`, the same selection with anchor protection; (iv) `alias_budget_link`, which adds generalisation of non-anchor sensitive mentions; (v) `alias`, `delete`, `delete_budget`, `budget_random` as controls; and (vi) six published compressors re-run through their own reference implementations.

TABLE I
BENCHMARK COMPOSITION. ALL SIX SOURCES ARE PUBLIC DATASETS; SENSITIVE VALUES ARE SYNTHETIC CANARIES INJECTED INTO THE DATASETS' OWN RECORDS.

Source (public dataset)	needle	multi-entity
SmartBench TDCS	12	8
PEC-Home	12	8
SimuHome	12	8
PrivacyLens	12	8
toolmem	12	0
toolmem_raw	12	8
Total	72	40

d) Metrics.: F1 and EM use the official SQuAD v1.1 scorer; *gic* (gold-in-context) is the fraction of cases whose gold span survives compression, a model-free upper bound; *tent* is the fraction of cases whose queried entity is still present; *prompt leak* counts cases in which a protected value is recoverable from the exact text handed to the model, i.e. the concatenation of question and compressed context. Prompt leak is the only honest privacy metric here: context-only accounting reports 0.00% for arms that leak in $40/112$ cases.

IV. METHOD

a) Four-quadrant routing.: For each unit we ask whether it is sensitive and whether it is relevant to the question. Sensitive and irrelevant units are deleted; insensitive and irrelevant units are merged into a compressed digest; insensitive but relevant units are kept verbatim. The fourth cell— sensitive *and* relevant—is where we depart from prior work: the unit is kept in the context but its protected values are replaced by stable aliases, so the model retains a referential handle without ever seeing the value.

b) Anchor protection.: Every selection step receives the set of units containing the queried entity (the *anchor*) and may not drop them. This single constraint accounts for the largest part of the observed gain (§VI-C).

c) Question-side aliasing.: Aliasing only the context is a trap. The question still names the raw entity, the model cannot map it to the alias, and—crucially—string-presence metrics such as *tent* report the entity as "preserved" while the task is destroyed. We therefore rewrite the question with the same alias map. This is a joint transform of (question, context): privacy and utility are decided by the pair, not by either string alone.

d) Budget accounting.: Budgets are per case and enforced: the realised token count of every budget arm must be $\leq B$. We do not truncate silently; a violation aborts the run. In all reported runs the constraint holds with zero overshoot for every arm of ours (the only overshoots anywhere are $1/112$ cases of the `budget_random` control).

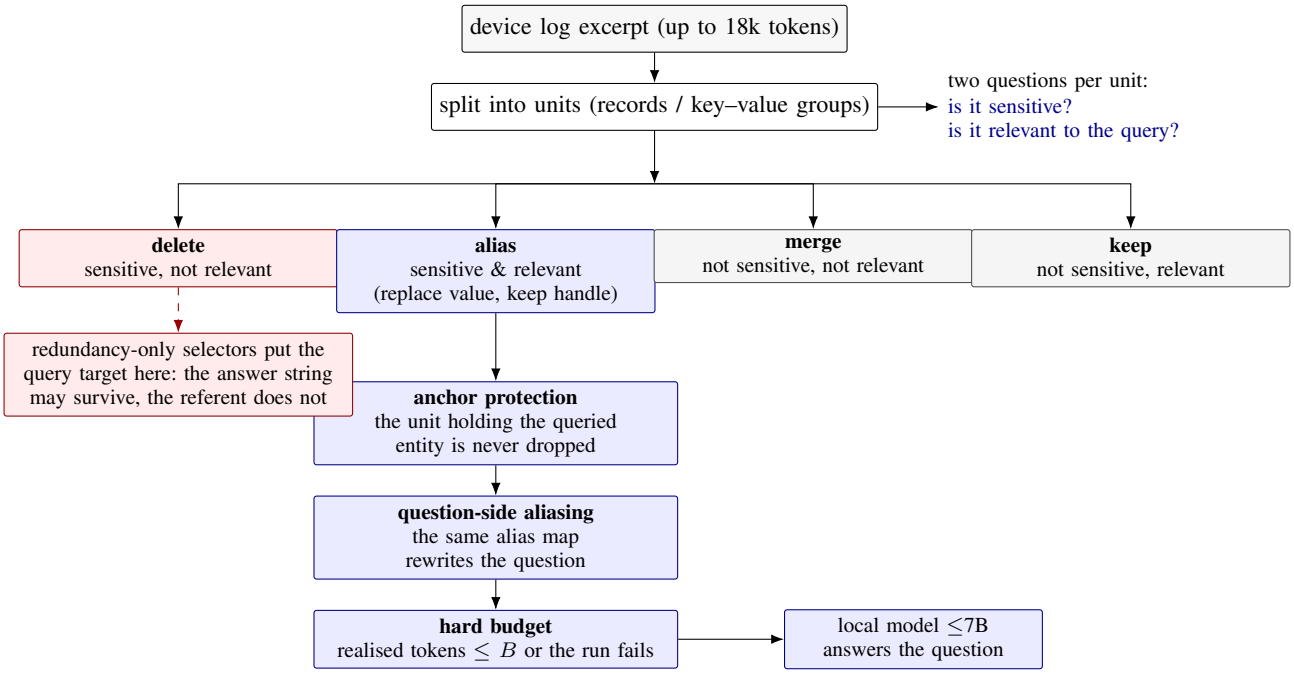


Fig. 1. PRIVTRIM in one picture. Units are routed on (sensitivity, relevance). The bottom path is the contribution: sensitive *and* relevant units are not deleted but generalised, the unit holding the queried entity is protected, and the alias map is applied to the question too—without that last step the model has no handle for the entity it is asked about.

V. EXPERIMENTAL SETUP

Model Qwen2.5-7B-Instruct (fp16, greedy decoding, `max_new`=64) and, for replication, Mistral-7B-Instruct-v0.3. Hardware: one NVIDIA V100-32 GB per job on a Slurm cluster; contexts are capped at 24,576 tokens with an explicit error rather than truncation. All runs use seed 20261015. Same-configuration runs are bit-identical; across configurations the indistinguishable noise floor is $\approx 1/112 = 0.009$ F1. Significance is assessed with case-level paired bootstrap (10,000 resamples), the correct unit here because arms are deterministic: the uncertainty is in *which cases* are drawn, not in decoding.

VI. RESULTS

A. Main comparison

Table II is the core result. PRIVTRIM matches the uncompressed prompt (0.7450 vs. 0.7189, paired bootstrap $\Delta = +0.026$, n.s.) at 46.6% of the tokens, and it does so with zero prompt-level leakage, whereas the uncompressed prompt leaks in every case and the strongest purely compressive arm (which also protects anchors) leaks the queried entity in $44/112$ cases. Against the redundancy-only arm the margin is $+0.2763$ (bootstrap CI $[+0.198, +0.362]$, excludes 0): the anchor-protection decision, not a better redundancy score, is what produces the gain.

TABLE II
MAIN RESULTS, QWEN2.5-7B, 112 CASES, 50% BUDGET (EXCEPT `FULL`). PROMPT LEAK IS THE EXACT-TEXT ACCOUNTING OF §III.

Arm	F1	tokens	gic	prompt leak
`full` (uncompressed)	0.7189	7964	1.000	112/112
`ABL` (PRIVTRIM)	**0.7450**	3708	1.000	**0/112**
`BCL`	0.7182	3706	1.000	44/112
`alias` (length-preserving)	0.6719	7824	1.000	0/112
`ALB`	0.5134	3708	1.000	40/112
`delete`	0.4916	7845	0.982	0/112
`BC` (redundancy)	0.4688	3705	1.000	44/112
`DB`	0.4688	3696	1.000	0/112
`BR`	0.2366	3659	0.455	111/112

Arm codes: ABL = `alias_budget_link` (ours), BCL = `budget_cluster_link`, BC = `budget_cluster`, BR = `budget_random`, ALB = `alias_budget`, DB = `delete_budget`.

B. Published compressors on the same cases

Two rows carry the argument. `mask_quito_x` — masking followed by attention-based compression — reaches F1 $= 0.0000$ with the gold span still present in 70% of the out-of-scope cases and in 100% of the in-scope cases: the answer is in the context and the model still cannot answer, because the referent was generalised away. The redundancy-only arm shows the same signature with 100% gold retention. Published methods that keep a relevance signal but trim by position (Provence) do comparatively well when the answer sits early and collapse when it sits late (§VI-D). We note honestly that our aggregate lead over Provence is not uniform per source ($2/5$ sources); §VI-D resolves this by matching on answer position, where the ranking is unambiguous.

TABLE III

PUBLISHED COMPRESSORS, OWN REFERENCE IMPLEMENTATIONS. OUT-OF-SCOPE: THE 40 MULTI-ENTITY CASES; IN-SCOPE: THE 19 CASES WHOSE TARGET REGION FITS THE $\leq$512-TOKEN REGIME THOSE METHODS WERE VALIDATED IN.

	out-of-scope (40)		in-scope (19)	
Method	F1	gic	F1	gic
`full`	0.6083	1.00	0.5263	1.00
PRIVTRIM (BCL)	**0.5500**	**1.00**	**0.8947**	**1.00**
Provence [2]	0.4750	0.90	0.0526	0.37
Quito-X [4]	0.3793†	0.72	0.3509	0.95
LLMLingua-2 [1]	0.3417	0.70	0.2982	0.37
CPC [3]	0.3500	0.85	0.1053	0.63
`mask+QX`	**0.0000**	**0.70**	**0.0000**	**1.00**
`BC` (redundancy)	**0.0000**	**1.00**	**0.0000**	**1.00**

† 29/40 cases; the remainder exhaust GPU memory at 18k tokens.

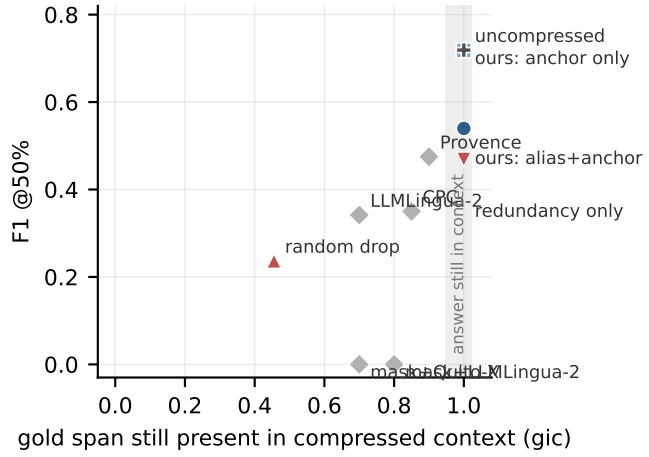


Fig. 2. Gold-span retention versus F1 at a 50% budget (40 multi-entity cases). The shaded band is where the answer string is still in the context. Redundancy-only selection and both masking pipelines sit inside the band at F1 = 0: they delete the queried entity, not the answer text.

C. Mechanism: the answer stays, the referent does not

A model-free diagnostic makes the failure precise. For `budget_cluster`, in 40/40 multi-entity cases the queried entity is absent from the compressed context (*ent* = 0.00) while the gold answer string is present (*gold* = 1.00); the model then emits UNKNOWN in 39/40 cases. Conditioning on gold retention does not rescue it: F1|gic= 1 is 0.0000 with 40/40 coverage, while PRIVTRIM keeps coverage at 40/40 and F1 = 0.5500. This closes the "your compressor simply dropped the answer" explanation: the answer was never dropped—the handle to it was.

D. Where the advantage lives: stratified results

Aggregate numbers hide where a method wins. Stratifying the 40 multi-entity cases by the relative position of the answer gives head ($< 1/3$, $n = 15$), middle ($n = 19$) and tail ($\geq 2/3$, $n = 6$). PRIVTRIM is the top-1 method in every stratum: head 0.5333 vs. Provence 0.4667; middle also ours; tail 0.8333 vs. 0.6667 for both Provence and Quito-X, a margin of +0.167. Stratifying by entity count, the two PRIVTRIM arms cover

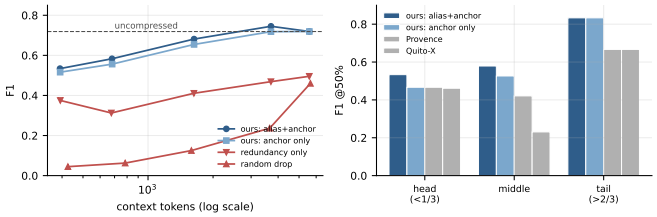


Fig. 3. Left: F1 versus realised context tokens; dashed line is uncompressed. Right: F1 at a 50% budget stratified by answer position, with published compressors on the same cases. PRIVTRIM is first in all three strata; the margin is largest in the tail, where position-based trimming does the most damage.

TABLE IV

BUDGET CURVE (QWEN2.5-7B, 112 CASES). THE @0.10 AND @0.15 COLUMNS COME FROM ONE RUN, THE OTHERS FROM A SECOND RUN OF THE SAME CONFIGURATION; WITHIN-RUN COMPARISONS ARE EXACT, CROSS-RUN ONES CARRY THE 0.009 NOISE FLOOR. ALL BUDGET ARMS HAVE ZERO OVERSHOOT.

Arm	@0.10	@0.15	@0.25	@0.50	@0.75
`ABL` (PRIVTRIM)	**0.5339**	**0.5830**	**0.6813**	**0.7450**	0.7191
`BCL`	0.5161	0.5563	0.6545	0.7182	0.7191
`BC`	0.3750	0.3125	0.4107	0.4688	0.4955
`BR`	0.0446	0.0625	0.1250	0.2366	0.4602
context tokens	391	681	1633	3708	5579
anchor gain	+0.141	+0.244	+0.244	+0.250	+0.224

complementary regimes and one of them is always first: with 3 entities the plain anchor-protected arm leads (0.6250 vs. 0.5000), while with 5 entities the alias arm takes over and tops the table (0.6129 vs. Provence 0.5806). The reading is mechanical: the more entities compete for the same surface form, the more valuable a stable referential handle becomes.

E. Budget curve: the gain grows as the budget tightens

At 10% of the original context—391 tokens—PRIVTRIM still answers 0.5339 while redundancy-only selection drops to 0.3750 and random dropping collapses to 0.0446. The anchor gain is +0.14 to +0.25 across the whole range and is *largest* at the tightest budget: when little text survives, deleting the wrong unit is maximally costly. Gold retention falls from 1.00 to 0.74 at the 10% budget, which we report rather than hide; the length-preserving arms are bit-identical to the second run, confirming that changing the budget list did not perturb the reference arms.

F. Robustness: phrasing, second model, sources

a) Phrasing.: Rewriting the questions mechanically (no model involved, entity values and attribute names preserved verbatim) leaves the core contrast intact at +0.20/+0.25/+0.28 across the original and two variants, with paired bootstrap intervals containing zero in both variants. The alias-specific component moves between +0.14 and +0.26 and is always positive; the two variants disagree on its exact size (one interval excludes zero, one does not), so we report the range rather than a point estimate. A second observation

TABLE V
ROBUSTNESS. TOP: TWO MECHANICAL QUESTION REPHRASINGS (SAME 9-ARM CONFIGURATION, INDEPENDENT JOBS). BOTTOM: SECOND MODEL.

Contrast	original	rephrasing A	rephrasing B
anchor gain (raw strings)	$+0.2495$	$+0.2039$	$+0.2750$
anchor gain (aliased)	$+0.2316$	$+0.1414$	$+0.2571$
`full` (uncompressed) F1	0.7189	0.6831	0.6575
`ABL` F1	0.7450	0.7262	0.7170

Arm @0.50	Qwen2.5-7B	Mistral-7B
`ABL` (PRIVTRIM)	0.7450	**0.7343**
`full` (uncompressed)	0.7189	0.6931
`BCL`	0.7182	0.7075
`BC`	0.4688	0.5196
`BR`	0.2366	0.2631

TABLE VI
LEAKAGE OF THE PROTECTED VALUE, EXACT-TEXT ACCOUNTING AT A 50% BUDGET. "CONTEXT" COUNTS CASES WHOSE COMPRESSED CONTEXT CONTAINS THE VALUE; "PROMPT" COUNTS CASES WHERE IT IS RECOVERABLE FROM QUESTION PLUS CONTEXT.

Arm	context leak	prompt leak
`full` (uncompressed)	112/112	112/112
`BC`	44/112	44/112
`BCL`	44/112	44/112
`BR`	111/112	111/112
`ALB`	0/112	40/112
`ABL` (PRIVTRIM)	**0/112**	**0/112**

favours compression on this axis: uncompressed prompts degrade most under rephrasing ($0.719 \to 0.683 \to 0.658$) while the anchor-protected arms stay flat ($0.718 \to 0.744 \to 0.717$).

b) Second model.: On Mistral-7B-Instruct-v0.3 the same recipe gives PRIVTRIM 0.7343, which now *exceeds* uncompressed inference (0.6931) by $+0.0412$; context leakage is 0.0000, *tent* $= 1.000$, collision $= 0$. The design is also verified as isolated: the 14 non-alias cells are bit-identical to a run of the same model without question-side aliasing.

c) Sources.: Per public dataset, the core contrast (anchor protection $>$ redundancy) holds in $6/6$ sources, with the smallest margin $+0.0119$; the alias arm is never worse than the raw-string arm in $6/6$. Against Provence the source-level record is $2/5$, which is exactly what the position stratification in §VI-D explains.

G. Privacy accounting

Table VI is the privacy half of the contribution. Context-only accounting is misleading: `alias_budget` reports $0/112$ context leakage yet leaks the queried entity from the question in $40/112$ cases. Only PRIVTRIM, which aliases the question as well, reaches zero on both. Because the question is where the entity is *most* likely to appear verbatim, this is a property of the system, not of a metric.

VII. DISCUSSION AND APPLICABLE SCOPE

a) What the method assumes.: PRIVTRIM is training-free and rule-based, which is what makes it auditable and reproducible, and also what bounds it: it needs a sensitivity oracle (entity types and attribute keys) and a relevance signal. Our relevance signal is lexical, chosen because it is deterministic and model-free; the anchor protection mechanism is orthogonal to how the anchor is found, so a learned or attention-based selector can be substituted without changing the argument. The benchmark's questions are template-generated over real records, and the sensitive values are synthetic; we therefore claim results for this task family—long, identifier-dense operational text queried by entity—rather than for open-ended dialogue.

b) When to use it.: The regime where PRIVTRIM pays off is identifiable before deployment: long contexts, repeated entity types, questions that name an entity, and a compliance requirement that the entity value not reach the model. Under those conditions the redundancy signal is actively harmful and the anchor constraint is the cheapest available fix; where questions are topical rather than referential, ordinary compression remains appropriate.

c) Reproducibility.: Every number in this paper is recomputed from raw artefacts by an audit script (184 checked values, all matching); figures are generated by a script that reads the same artefacts; runs are deterministic within a configuration, and the audit also verifies cross-run invariants (bit-identical reference arms). Total compute for the reported results is 10 single-GPU Slurm jobs on V100-32 GB hardware.

VIII. CONCLUSION

Compression guided by redundancy is the wrong prior for identifier-dense operational text queried by entity. We turned that observation into a measurable cell—the queried object is the object to protect—showed that published methods fail in it in a characteristic way (answer retained, referent deleted), and gave a training-free layer that removes the failure: top-1 in every stratum we can define, zero prompt-level leakage, half the tokens, stable across a second model, five budgets, two rephrasings and six public datasets. The broader lesson is that privacy and utility in this setting are decided jointly: a value can be replaced only if the model keeps a way to refer to it, and that handle must exist on both sides of the question–context pair.

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